Sampling and Data

Anaya Scott

Question 1

1/1 pt 2 19

Determine whether the value is a parameter or statistic

43.87% of voters turned out for the 2004 elections

- Statistic
- Parameter

Question Help: Read

Question 2

1/1 pt 2 19

In a study, the data you collect is have you ever done binge drinking in college.

What type of data is this?

- qualitative (categorical)
- quantitative

Question Help: Video

Question 3

1/1 pt 1 19

A political scientist surveys 39 of the current 138 representatives in a state's legislature.

What is the size of the sample: 39

What is the size of the population: 138

Question Help: Video 1 Video 2

Question 4

1/1 pt 2 19

A team of researchers is testing the effectiveness of a new HPV vaccine. They randomly divide the subjects into two groups.

Group 1 receives new HPV vaccine, and Group 2 receives existing HPV vaccine.

The patients in the study do not know which group they are in.

Which is the treatment group?

- Group 1
- Group 2
- Neither group

Which is the control group (if there is one)?

- Group 1
- Group 2
- No control group

Is this study blind, double blind, or neither?

- Blind
- Double-blind
- Neither

Question Help: Video

Question 5 1/1 pt 2 19

Suppose you want to know what high school incoming freshman at a particular college came from. To estimate the percentage of incoming freshman from each of the area's high schools you take a sample of 200 incoming college freshman.

Match the following:

- e an incoming college freshman
- d 200 incoming college freshman
- c percentage of all incoming freshman who graduated from each area high school
- b percentage of the 200 incoming college freshmen who graduated from each area high school
- f high school graduated from
- a all incoming college freshmen

a. population

b. statistic

c. parameter

d. sample

e. individual

f. variable

Question Help: Video

✓ **Question 6** 1/1 pt 2 19

A political scientist surveys 29 of the current 65 representatives in a state's legislature.

What is the size of the sample: 29

What is the size of the population: 65

Question Help: Video 1 Video 2

✓ **Question 7** 1/1 pt 1 19

To test a new teaching method for reading first grade students are randomly assigned to either a teacher teaching using the old method or a teacher using the new method.

Is this an observational study or an experiment?

- experiment
- observational study

Question Help: Video 1 Video 2

Question 8

1/1 pt 2 19

A researcher wants to determine if a new drug lowers the blood pressure of patients with high blood pressure. The patients are randomly selected to be in the study and then randomly assigned to a group.

Is this a randomized experiment? Why or why not?

- yes, since the treatment is chosen by the person in the study
- no, since the treatment is not randomly selected by the researcher
- yes, since the treatment is randomly selected by the researcher

Question Help: Video

Question 9

1/1 pt 1 19

A business manager wants to see if a new procedure improves the processing time for a task. The manager measures the processing time of a random sample of employees using the old procedure and compares it to the processing time of a random sample of employees using the new system.

Are the two samples matched pairs? Why or why not?

- yes, two measurements of the same variable have been taken on the same object
- yes, two measurements of the same variable have been taken on different objects that have been matched based on some criteria
- no, measurements of two different variables were taken on each object
- no, two measurements of the same variable have been taken on the same object
- no, two measurements of the same variable have not been taken on the same object or matched objects

Question Help: Video

Question 10

1/1 pt 2 19

To determine if a new medication reduces headache pain, some patients are given the new medication and others are given a placebo. Neither the researchers nor the patients know who is taking the real medication and who is taking the placebo.

Is this a blind experiment, double blind experiment, or neither? Why?

- neither, since both the researcher and the individual know who is getting which treatment
- double blind, since neither the researcher nor the individual knows who is getting which treatment
- blind, since neither the researcher nor the individual knows who is getting which treatment
- blind, since the researcher knows who is getting which treatment but the individual does not
- double blind, since both the researcher and the individual know who is getting which treatment
- blind, since the individual knows who is getting which treatment but the researcher does not

✓

Question Help: Video

✓ Question 11

1/1 pt 2 19

In a study, the data you collect is Weight in pounds.

This data is:

- Quantitative
- Categorical

✓

✓ Question 12

1/1 pt 1 19

Classify the data as qualitative or quantitative. If qualitative then classify it as ordinal or categorical, and if quantitative then classify it as discrete or continuous.

Variable	Data Type
d Shoe Size	a. Qualitative Ordinal
c Weight (exact)	b. Qualitative Categorical
a Grade (ABCDF)	c. Quantitative Continuous
b Gender	d. Quantitative Discrete

✓

Question Help: Video

✓ Question 13

1/1 pt 2 19

Classify a statistical study as either an observational or a designed experiment.

Statistical Study

Designed Experiment In a clinical trial, 780 subparticipants suffering from high blood pressure were randomly assigned to one of three groups. Over a one-month period, the first group received the experimental drug, the second group received a placebo, and the third group received no treatment. The diastolic blood pressure of each participant was measured at the beginning and at the end of the period and the change in blood pressure was recorded. The average change in blood pressure was calculated foreach of the three groups and the three averages were compared.

Observational Study An educational researcher used school records to determine that, in one school district,84% of children living in two-parent homes graduated high school while 75% of children living insingle-parent homes graduated high school.

Question Help: Video

Question 14 1/1 pt 2 19

Match the key term with each item listed below.

A cardiologist is interested in the mean recovery period of her patients who have had heart attacks.

Item

c a group of patients who have had heart attacks randomly selected

e the recovery time

b the average recovery time of the randomly selected group of patients who had a heart attack

d all patients who have had heart attacks

f the recovery time of one person who had a heart attack

a the average recovery time of all people who have heart attacks

Key Term

a. parameter
b. statistic
c. sample
d. population
e. variable
f. data

✓

Question 15 1/1 pt 2 19

Choose the appropriate term for each definition below.

mean the sum of the observations divided by the number of observations

mode the most frequently occurring value in the data set

range the difference between the maximum (largest) and minimum (smallest) observations

z-score the number of standard deviations an observation is away from the mean

second quartile (Q2) a number for which 50% of the data is less than that number; same as the median

✓

Question Help: Read

Question 16

1/1 pt 2 19

Choose the appropriate term for each definition below.

Binomial random variable a discrete random variable that represents the number of successes among n Bernoulli trials.

Continuous random variable a random variable whose possible values form a continuous interval

Binomial experiment an experiment in which there are exactly two possible outcomes - success and failure, with the probability of success $P(S)=p$ and the probability of failure $P(F)=1-p$.

Random variable an unknown quantity whose value depends on chance

Probability distribution table the table that summarizes the possible values of a discrete random variable and the corresponding probabilities

Question Help: Read

Question 17

1/1 pt 1 19

What is the definition of a simple random sample?

- A simple random sampling is a sample of size n from a population N where each person is guaranteed to be chosen at least once.
- A simple random sampling is a sample of size n from a population N where each sample of size n has an equal chance of being chosen.
- A simple random sampling is a sample of size n from a population n where each person is guaranteed to be chosen.
- A simple random sampling is a convenient sample of size n from a population N.

Question Help: Video

Question 18

1/1 pt 2 19

Choose the appropriate term for each definition below.

hypothesis test a test to decide whether the null hypothesis should be rejected in favor of the alternative hypothesis or not

statistical claim a statement about a population parameter in the form of an equation or an inequality

type 1 error rejecting a true claim

alternative hypothesis a statistical claim in the form of an inequality to be tested as an alternative to the null hypothesis

type 2 error failing to reject a false claim

✓

Question Help: Read

✓ **Question 19** 1/1 pt 2 19

Choose the appropriate term for each definition below.

independent samples a pair of samples in which the observations in one sample do not influence the observations in the other

pooled standard deviation the standard deviation computed by treating the two samples as one

pooled proportion the proportion obtained by treating the two samples as one

paired samples a pair of samples in which the observations are paired in a distinct way

dependent samples a pair of samples in which the observations in one sample somehow influence the observations in the other

✓

Question Help: Read

✓ **Question 20** 1/1 pt 2 19

Choose the appropriate term for each definition below.

the Central Limit Theorem for Sample Sums for sufficiently large samples (n>30), the distribution of the sample sum is approximately normal

the Central Limit Theorem for Sample Proportions for sufficiently large samples (np>10 and n(1-p)>10), the distribution of the sample proportion is approximately normal

the sample variance a random variable whose possible values are the variances of the samples of size n from the population

the sample proportion a random variable that represents the number of cases falling into one category of the variable divided by the number of cases in the sample

the sample mean a random variable whose values are the averages of the samples of size n from the population

Question Help: Read

Question 21

1/1 pt 2 19

A pharmaceutical company created a drug that is supposed to lower the risk of heart disease. After a series of clinical trials on 32 patients it was determined that it lowers the risk of heart disease by 3% and the cost of production is 68 dollars per dose. Using methods of statistics it can be shown that this result is unlikely to happen due to chance even if the drug has no effect.

Do the drug trials have statistical significance? Yes

Do drug trials have practical significance? No

Show Detailed Solution

Question Help: Video

Question 22

1/1 pt 2 19

Recall that correlation doesn't imply causation! In each scenario below, determine the lurking variable.

Correlation Scenario

Temperature — A study finds that glove sales and snowboarding accidents are highly correlated.

Age — The height of an elementary school student and his or her reading level.

Cloud cover — After the war statisticians studied the accuracy of strategic bombing. Some things made sense (different types of bombers had different accuracy levels, higher altitude meant less accuracy). But one variable was whether enemy fighters opposed the bombers, and this had the 'opposite' effect - fighter opposition meant more accuracy.

✓

Question Help: Video

✓ Question 23

1/1 pt 2 19

Classify each of the sampling procedures below.

Sampling Procedure

Cluster Sampling — A pollster interviews all human resource personnel in five different high tech companies.

Stratified Sampling — A soccer coach selects six players from a group of boys aged eight to ten, seven players from a group of boys aged 11 to 12, and three players from a group of boys aged 13 to 14 to form a recreational soccer team.

Systematic Sampling — A medical researcher interviews every third cancer patient from a list of cancer patients at a local hospital.

Simple Random Sampling — A high school counselor uses a computer to generate 50 random numbers and then picks students whose names correspond to the numbers.

✓

Question Help: Video 1 Video 2 Video 3 Video 4

✓ Question 24

1/1 pt 2 19

Determine whether the sampling method described below appears to be sound or is flawed.

In a survey of 114 subjects, each was asked how often he or she read a newspaper. The survey subjects were internet users who responded to a question that was posted on a news website.

- ○ It appears to be sound because it is a census.
- ◉ It is flawed because it is a voluntary response sample.
- ○ It appears to be sound because the data are not biased in any way.
- ○ It is flawed because it is from a small sample.

✓

Question Help: Video

Question 25

1/1 pt 2 19

Match each variable with its Level of Measurement.

d	Letter Grade (A, B, C, D, or F)	a. Interval
c	Wage per hour	b. Nominal
c	Age in years	c. Ratio
b	Gender	d. Ordinal
d	Habits on an Always/Sometimes/Never scale	
a	Temperature in Celsius	

Question Help: Video

Question 26

1/1 pt 2 19

Determine whether the following data sets are best described as: Nominal, Ordinal, Interval or Ratio. If it is ratio or interval, also tell whether it is discrete or continuous.

a. The radius of ball bearings manufactured by a company.
Ratio and Continuous

b. The amount of candy a child receives at Halloween.
Ratio and Discrete

c. The names of constellations.
Nominal

d. The months of the year.
Ordinal

Question Help: Video

Question 27

1/1 pt 2 19

To test a new teaching method for reading first grade students are randomly assigned to either a teacher teaching using the old method or a teacher using the new method.

- observational study
- experiment

✓

Question Help: Video 1 Video 2

✓ Question 28

1/1 pt 2 19

Match each of the following to the type of sampling technique used:

c A school district has a list of all of its 9th grade students. They then collect data on the PSAT score of 30 randomly selected students.

d The school district divides the 9th grade students by high school attended and then collects data from a random sample of 9th graders from each high school.

a The school district randomly selects 3 high schools in the district and then collects data on PSAT score for all 9th graders in those districts.

e The school district collects data on the PSAT score of the first 50 scores that are entered into the system.

b The school district has a list of all 9th grade students and then collects data on the PSAT score of every 10th student on the list.

a. cluster sample
b. systematic sample
c. simple random sample
d. stratified sample
e. convenience sample

✓

Question Help: Video

✓ Question 29

1/1 pt 2 19

After giving a test to a group of students, below are the summary of grades and gender. Write as a reduced fraction.

	A	B	C	Total
Male	9	20	12	41
Female	15	4	7	26
Total	24	24	19	67

If a student is randomly chosen, find the probability that the student did *not* earn an "A"

$\frac{43}{67}$

Question Help: Video